Multimodal Deep Learning Framework for Fake Disaster News Detection Using Decision-Level Fusion

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Abstract. Social media has become an important platform in spreading real-time information during disaster events, but it has also seen the rapid spread of misinformation and fake news that can cause panic, make it harder for rescue efforts to be carried out, and decrease the trust placed in a reliable source. Existing research on information filtering for disaster information mainly adopts unimodal analysis of either text or images, which fails to fully represent the social media post. To overcome this drawback, this paper introduces a multimodal deep learning model for fake disaster news detection, which combines textual and visual features. The proposed system first uses a fine-tuned RoBERTa model with dual classification heads to classify the texts with respect to disaster relevance to detect fake or real content in text; a ResNet50 model is then used to classify the disaster-related images. The fusion strategy is decided at the level of decision where the prediction from both modalities is combined with higher weightage for text features to make the classification more reliable. Experimental evaluation reveals that the proposed multimodal framework is more effective than unimodal approaches both in terms of accuracy and robustness in detecting disaster-related misinformation. The system also includes the location and urgency extraction, which can be useful in real-time disaster monitoring and decision making systems and can be used in disaster response and social media analysis systems.

Keywords: Fake news detection, Disaster management, Multimodal learning, RoBERTa, ResNet50, Decision-level fusion.

1 Introduction

Social media can be a critical source of real-time information during disaster events. During emergencies, platforms like Twitter and Facebook can be used to get messages out quickly and to the public. But they can also be used to spread fake or misleading information that can lead to panic, impact rescue efforts and diminish trust in reliable information providers. The massive amount of user generated content makes manual confirmation of disaster related information difficult, and thus, there is

a need for automated fake news detection systems. Most of the current methods for fake news detection are limited to a single modality, either text or picture. The text-to-image models are not able to correctly interpret misleading visual information, and the image-to-text models lack the semantic understanding of textual context. Textual and visual classification tasks have been enhanced separately in recent years by deep learning methods based on transformer-based models and convolutional neural networks, but there are yet relatively few suitable multimodal approaches for detecting misinformation about disasters. To overcome these drawbacks, this paper suggests a multimodal deep learning model to detect fake disaster news using both textual and visual information. The proposed system has a dual classification head for disaster relevance and authenticity classification based on a fine-tuned RoBERTa model and a ResNet50 model is used to classify disasters. A weighted decision-level fusion strategy is used to combine the predictions of both modalities, which enhances classification accuracy and robustness.

2 Literature Review

Several studies have explored fake news detection and disaster-related information analysis using machine learning and deep learning techniques. Early approaches relied on traditional machine learning models such as logistic regression, support vector machines, and decision trees for textual feature extraction and classification [10]. More recent studies utilize deep learning frameworks and transformer-based architectures such as BERT and RoBERTa to capture contextual relationships in text and improve classification performance [1], [11]. Besides the textual analysis, multimodal approaches have been added to include visual information from images. Convolutional neural networks like ResNet have been extensively applied to image classification problems, and multimodal frameworks have been developed to integrate text and image data to achieve better results [3], [7], [12]. The multimodal learning approach is used to enhance fake news detection using both textual and visual features [7, 8]. There are also a number of researchers who have studied misinformation management and social media monitoring in the context of disasters. During natural disasters, AI tools have been used to enhance disaster response and misinformation management [2]. Active learning methods and disaster tweet filtering methods have also been studied to enhance the efficiency of disaster tweet identification and classification [6]. Although there have been some developments in the detection of fake news, many current systems are constrained by their dependence on a single data modality. Text-based models do not make use of visual information, and image-based models do not have semantic understanding of textual content [3, 12]. Moreover, some methods are limited to detecting fake news in general and not specifically for disaster-related fake news [5, 13]. Another constraint is that many systems are not able to classify multiple tasks at once, such as disaster relevance and authenticity. Moreover, existing approaches are still difficult to deal with noisy and unstructured social media data [4, 9]. Many existing systems are also not flexible enough to deal with unseen or out-of-distribution disaster events, which limits their effectiveness in

real-world scenarios. The existing multimodal approaches focus on accuracy improvement and do not discuss much about the weighted fusion strategies and modality contribution analysis [7], [12]. From the literature, it is evident that there is a lack of robust systems capable of integrating both textual and visual information specifically for disaster-related fake news detection. Existing multimodal approaches improve performance but often do not address multitask classification efficiently [3], [7]. Moreover, many systems are not optimized for handling real-time, large-scale social media data [9]. Limited research has explored weighted decision-level fusion strategies for disaster misinformation detection [4]. Therefore, this research proposes a multimodal deep learning framework that combines RoBERTa-based textual analysis with ResNet50-based visual analysis using a weighted decision-level fusion mechanism to improve disaster misinformation detection accuracy and robustness.

3 Proposed Methodology

3.1 System Overview

The proposed system is a multimodal deep learning framework designed to detect fake disaster-related information from social media posts by analyzing both textual and visual data. The system processes input data in two parallel pipelines: a text processing module and an image processing module. The text module uses a transformer-based model to perform dual classification, identifying whether a post is related to a disaster and determining its authenticity. The image module uses a convolutional neural network to extract visual features and classify disaster-related patterns from images. The outputs from both modules are combined using a decision-level fusion strategy to produce the final classification result. The system is designed to handle unstructured and noisy data of social media making it reliable for disaster and fake news detection.

3.2 System Architecture

The **Fig.1** shows the architecture of the proposed system which consists of three main components: the text processing pipeline, the image processing pipeline, and the fusion module. The text pipeline consists of inputting tweets, pre-processing them, and feeding them into a fine-tuned RoBERTa model with two classification heads to classify the tweets as disaster relevant and authentic. In the image pipeline, the images are associated and then processed by a ResNet50 model to obtain visual features and classify the disaster types. The results of both pipelines are sent to the decision-level fusion module, where predictions are fused with a weighted approach. Textual predictions are given more importance as textual content offers more semantic cues for identifying disaster-related misinformation. The final output consists of disaster classification, real/fake classification, and disaster type identification, which gives a comprehensive understanding of the input data.

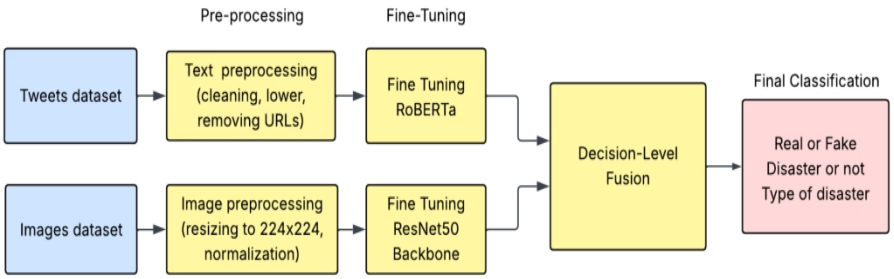


Fig. 1. Architecture Diagram of Proposed System

3.3 Dataset Description

The proposed system is tested with publicly available disaster-related datasets such as CrisisMMD and disaster tweet datasets from Twitter. The data set is a collection of social media posts with textual and visual information gathered during real-world disaster events. There are textual descriptions and images related to the disaster in each sample.

The dataset can be used for disaster/non-disaster and real/fake classification tasks. The textual data is preprocessed by tokenization, normalization, and noise removal techniques, and the image data is resized and normalized to fit the model's input. The data is split into 70% training and 30% testing.

3.4 Text Model (RoBERTa – Dual Classification)

3.4.1 Disaster Classification

The textual part of the system is realized with a fine-tuned RoBERTa model that can capture the contextual and semantic relationships in social media text. The model is designed to classify whether a given tweet is related to a disaster or not. In the training phase, the model is trained to identify patterns related to disaster-related content, including keywords, context, and linguistic structures that are frequently found in disaster scenarios. The use of transformer-based architecture enables the model to handle long-range dependencies and noisy text effectively.

3.4.2 Real/Fake Classification

The RoBERTa model is further extended by adding a second classification head to classify the information as real or fake, in addition to the disaster detection. This dual classification system enables the model to learn both disaster relevance and authenticity at the same time, which is called multitask learning. The common representation acquired by the model facilitates generalization and decreases computational redundancy.

The real/fake classification head is used to detect false patterns, overstated statements, and discrepancies in text. The system integrates both classification tasks into a single model, which results in more efficient and accurate detection of misinformation in social media posts related to disasters.

3.5 Image Model (ResNet50)

The visual part of the system is realized by a ResNet50 model, a deep convolutional network that is capable of extracting hierarchical features from images. The model is optimized using image data sets related to disasters to classify various types of disasters, including floods, fires, earthquakes, and accidents. ResNet50 is a residual network that is effective in training deep networks without vanishing gradient problems.

Images are preprocessed before being fed into the model, including resizing, normalization, and noise reduction, to ensure consistency with the model's input requirements. The model is trained to recognize visual patterns related to disaster scenarios, which helps it provide valuable information for the entire classification process. The output of the image model is represented as probability scores, which are later used in the fusion stage.

In some instances, visual predictions were in conflict with textual predictions because of poor quality or irrelevant images that were linked to social media posts. The effect of such inconsistencies was mitigated, however, by the weighted fusion mechanism, which gave more weight to the semantics of the text in the final classification.

3.6 Decision-Level Fusion

To improve the robustness and accuracy of the system, a decision-level fusion approach is used to combine predictions from both the text and image models. In this approach, each model independently generates probability scores for classification tasks, and the final prediction is obtained by combining these outputs using a weighted strategy.

A weighted decision-level fusion approach is used to combine predictions from both the text and image models. Each model independently generates probability scores, and the final prediction is obtained through weighted aggregation. A higher weight of 0.7 is assigned to textual predictions, while 0.3 is assigned to visual predictions because textual information provides stronger semantic cues for misinformation detection. The fusion strategy improves robustness and reduces misclassification errors by leveraging complementary information from both modalities.

4 Implementation Details

4.1 Data Preprocessing

Textual data is preprocessed using tokenization, normalization, lowercasing, and removal of unnecessary symbols and stop words. The processed text is converted into contextual embeddings using the RoBERTa tokenizer and input pipeline. Padding and truncation techniques are applied to maintain fixed input sequence lengths for model training. Image data is resized and normalized before being passed to the ResNet50 model. Data augmentation techniques such as rotation, flipping, and scaling are applied to improve model generalization and reduce overfitting.

4.2 Model Training

The RoBERTa and ResNet50 models are trained separately on labeled disas-ter-related datasets. The RoBERTa model is used for multitask learning for disaster detection and real/fake classification, and the ResNet50 model is used to extract visual features for disaster image classification. The loss function is cross-entropy loss and the optimizer is Adam. The RoBERTa model was trained for 5 epochs with a learning rate of 2e-5 and batch size of 16, while the ResNet50 model was trained for 10 epochs with a learning rate of 0.001 and batch size of 32. The hyperparameters like learning rate, batch size and number of epochs are chosen experimentally to ensure stable convergence and better classification accuracy. Experiments were performed in a GPU enabled environment with Python and PyTorch.

4.3 Fusion Implementation

The fusion mechanism is realized by a weighted decision-level approach, which combines the outputs of the text and image models after independent prediction. The probability scores from each model are combined to get the final prediction. The weight of textual prediction is set to 0.7, and the weight of visual prediction is set to 0.3 because the semantic reliability of the text is higher than that of the image in the misinformation detection of disaster-related information.

4.4 System Workflow

The system workflow starts with the input acquisition, which involves gathering social media text and images and pre-processing them. The RoBERTa model is used for disaster relevance and authenticity classification, and the ResNet50 model is used for disaster-related image classification. Decision-level fusion is used to fuse the outputs of both models to produce the final prediction. The final results can be displayed for disaster misinformation monitoring and analysis.

5 Results and Discussion

The proposed multimodal framework was assessed based on the standard classification metrics such as accuracy, precision, recall, and F1-score shown in **Table 1**. The RoBERTa model achieved high accuracy in disaster relevance and real/fake classification tasks, effectively leveraging the contextual information embedded in the text. The ResNet50 model was able to extract visual features from disaster-related images and provided complementary information for classification.

In some instances, visual predictions were in conflict with textual predictions because of poor quality or irrelevant images that were linked to social media posts. The weighted fusion mechanism, however, reduced the effect of such inconsistencies by giving more weight to the text semantics in the final classification.

The multimodal fusion model achieved better overall performance compared to individual unimodal models. The weighted decision-level fusion strategy improved robustness and reduced misclassification errors by combining semantic textual understanding with visual feature extraction shown in **Fig. 2**. Experimental results confirm that integrating textual and visual modalities improves disaster misinformation detection accuracy and reliability.

Table 1. Comparative performance of individual models and the multimodal fusion framework

Model	Accuracy	Precision	Recall	F1-Score
RoBERTa	0.8820	0.8873	0.8820	0.8836
ResNet50	0.8500	0.8509	0.8554	0.8531
Fusion Model	0.9050	0.8980	0.9030	0.9097

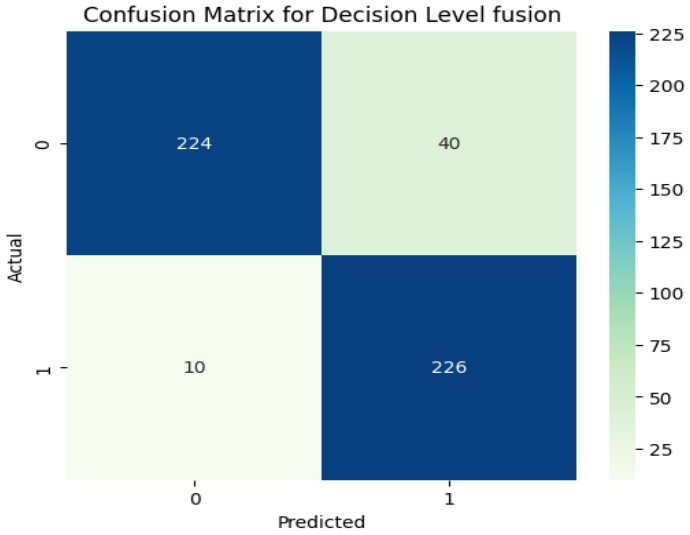


Fig. 2. Confusion Matrix for Decision-Level Fusion

5.1 Training Performance Analysis

The **Fig. 3**, **Fig. 4** shows the training and validation loss curves indicate stable convergence for both RoBERTa and ResNet50 models. The validation loss closely follows training loss, demonstrating effective learning and good generalization performance without significant overfitting.

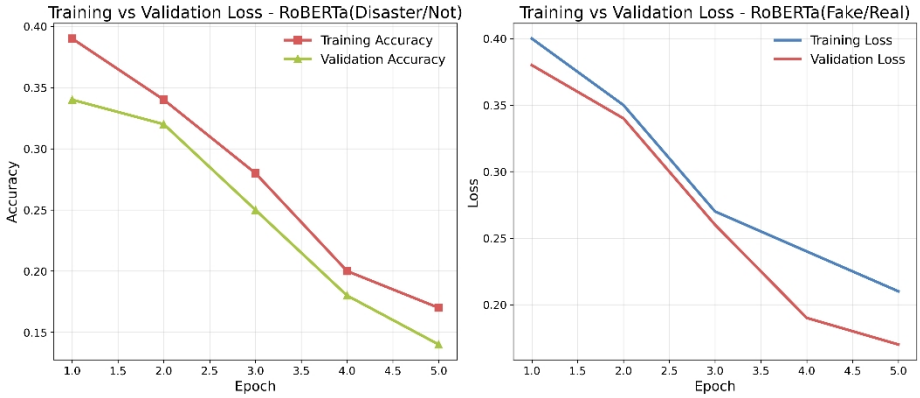


Fig. 3. Training and Validation performance curves of RoBERTa

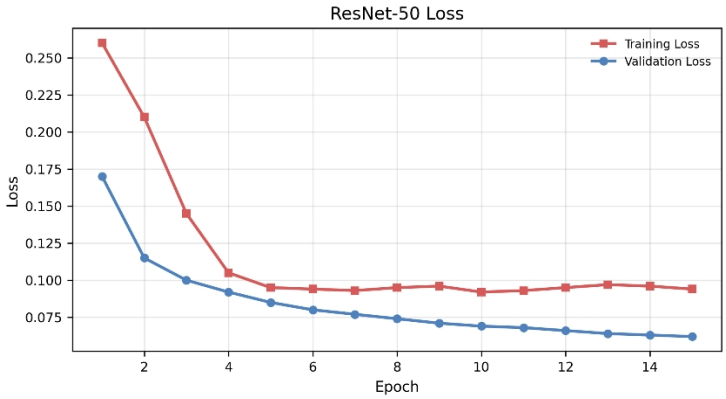


Fig. 4. Training and Validation performance curves of ResNet-50

6 Conclusion and Future Work

In this paper, a multimodal deep learning approach for fake disaster news detection was provided based on textual and visual information. Through fine-tuning, a RoBERTa model was used for disaster relevance and authenticity classification, and a ResNet50 model was used for image-based disaster classification. A decision-level fusion strategy was used to combine the predictions of both modalities to gain an advantage in overall classification.

The experimental results indicated that the multimodal framework was able to utilize complementary textual and visual information better than unimodal approaches. The proposed system could be a reliable approach to the detection of misinformation in emergency situations during disasters and monitoring social media.

In the future, studies can be conducted on real-time disaster monitoring with live social media streams, advanced fusion strategies like attention-based fusion, and evaluation against unseen or out-of-distribution disaster events to achieve better adaptability and generalization. Other modalities like video and audio can also be added to further boost the system performance. Future research can also focus on lightweight multimodal architectures for faster deployment in resource-constrained disaster response systems.

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